Comprehensive Strategic and Algorithmic Audit: Advancing Galaxy Sports Edge (GSE) Prediction Infrastructure

The Strategic Imperative and Systemic Vulnerabilities

The consolidation of the sports prediction and affiliate media industry has created a structural vacuum that Galaxy Sports Edge (GSE) is uniquely positioned to fill. An analysis of the competitor landscape reveals that the traditional affiliate model is experiencing a severe decline. Conglomerates such as Better Collective, which operates the Action Network, are facing compounding headwinds including the disintermediation of traffic by artificial intelligence chatbots, the rapid rise of prediction markets like Kalshi and Polymarket that do not pay affiliate cost-per-action (CPA) fees, and increasingly stringent regulatory environments. Concurrently, the Federal Trade Commission (FTC) has signaled a zero-tolerance approach to unsubstantiated earnings claims, evidenced by the $2.425 million settlement levied against the financial subscription service RagingBull.com for deceptive win-rate marketing. Furthermore, the daily fantasy sports sector is navigating existential legal threats, with operators like PrizePicks and Underdog facing respective $15 million and $17.5 million fines for offering house-backed player propositions that regulators classify as illegal lotteries.

Within this precarious ecosystem, GSE's commitment to an honesty-gated "Glass Ledger"—a cryptographically timestamped, independently verifiable record of closing-line value (CLV) and realized accuracy—serves as an impenetrable legal and commercial moat. However, an exhaustive audit of GSE's internal production receipts (model version v5.2.7) exposes critical theoretical and architectural blind spots that threaten the integrity of this ledger. The live record audit of 1,111 receipts reveals a severe data-integrity flaw, with 17.9% of rows poisoned by invalid American odds inputs, rendering naive unit-based return on investment (ROI) calculations effectively meaningless. More alarmingly, the production data exhibits a pronounced "confidence inversion." Rather than scaling linearly, realized win rates precipitate a sharp decline as model confidence increases. For instance, predictions registering a 50-54% confidence level achieved a 57.1% realized win rate, whereas those categorized at 80% confidence or higher plummeted to a 37.5% win rate.

This confidence inversion is a textbook mathematical consequence of relying on global, marginal calibration techniques over highly heterogeneous data. The current architecture employs global isotonic regression and temperature scaling, which minimize the global Expected Calibration Error (ECE) but invariably trigger minority collapse. By pooling disparate market regimes—such as heavily juiced favorites, volatile longshots, and distinct weather paradigms—into a single calibration curve, the system artificially suppresses the variance of minority subpopulations to satisfy a macro-level objective function.

To resolve these vulnerabilities and elevate the predictive infrastructure beyond incumbent methodologies, this report outlines a comprehensive, falsifiable research program. It introduces advanced conformal risk control, copula-based dependence structures, continuous-time survival analysis, bipolar decision theory under Knightian uncertainty, and causal inference techniques, all rigorously mapped to the existing Beexly/Sports repository.

Conformal Risk Control and Mondrian Calibration Strategies

The reliance on a global ECE threshold of 0.06 over 250 settled samples, while a commendable foundational honesty gate, is mathematically insufficient for a product whose core value proposition hinges on local predictive edge. To rectify the confidence inversion and eliminate minority collapse, the infrastructure must abandon purely marginal calibration in favor of Mondrian Conformal Prediction and Conformal Risk Control (CRC).

Mondrian Conformal Prediction addresses the heterogeneity of the feature space by partitioning the data into predefined, non-overlapping strata, or "Mondrian categories," based on domain-specific covariates such as sport, market type, and bookmaker disagreement levels. By computing separate calibration quantiles within each stratum, the system guarantees conditional coverage, ensuring that a 90% prediction interval maintains its validity specifically for NFL passing props in high-wind conditions, rather than merely averaging out against highly predictable moneyline favorites.

Expanding upon this, the system must integrate Conformal Risk Control, a theoretical framework pioneered by Angelopoulos, Bates, and colleagues, which extends the conformal methodology beyond simple miscoverage to bound the expected value of any monotonically decreasing loss function. In the context of GSE's operational architecture, the relevant loss function is the rate of unprofitable or severely miscalibrated posted slates. By utilizing a holdout calibration set, the CRC algorithm identifies a critical non-conformity threshold. It leverages finite-sample concentration inequalities, such as Hoeffding or Bentkus bounds, to ensure that the empirical risk remains strictly below a user-defined tolerance level α.

This mechanism serves as a mathematically guaranteed "sliding rheostat" for the gse-score/calibration-contract.ts module. Instead of relying on ad-hoc heuristic vetoes, CRC provides a formal loss-rate contract. If the required threshold to maintain the risk budget results in an empty prediction set for a specific input, the system deterministically triggers a HARD_PASS (No-Bet) decision. Furthermore, this approach mitigates the score-mechanism shift observed in multi-agent LLM systems, where peer pressure or market consensus can silently degrade conformal coverage by artificially altering the softmax probabilities of correct answers. To supplement point estimates, the integration of Venn-Abers predictors—which utilize isotonic regression across calibration folds to output a multiprobability interval—will provide a robust, finite-sample valid range of uncertainty that can be surfaced directly to the user, fulfilling the mandate for transparent, calibrated displays.

Advanced Statistical Dependence: Copulas and Joint Modeling

A systemic vulnerability in the current poisson.ts and dixon-coles.ts implementations is the assumption of either strict independence or highly constrained positive correlation between discrete events. In modern sports betting markets, particularly within the highly lucrative Same-Game Parlay (SGP) segment, outcomes are inextricably linked by complex, often asymmetric dependence structures. Treating a team's total points and its starting quarterback's passing yards as independent variables leads to severe mispricing and exposes the platform to adverse selection.

To model these dynamics accurately, the prediction engine must adopt the Sarmanov Conway-Maxwell-Poisson (COM-Poisson) distribution. The marginal COM-Poisson distribution introduces a vital dispersion parameter (ν) to the traditional Poisson rate (λ). This allows the model to seamlessly handle equidispersed ($\nu=1$), overdispersed ($\nu<1$), and underdispersed ($\nu>1$) count data, which is essential for capturing the variance in scoring rates across different offensive philosophies.

The critical innovation lies in coupling these marginals using a Sarmanov mixing kernel. Traditional bivariate Poisson constructions generate dependence through a shared latent variable, which mathematically restricts the correlation to non-negative values. The Sarmanov methodology utilizes bounded mixing functions to construct a joint probability mass function capable of capturing the full spectrum of correlation, including the negative dependence frequently observed during "garbage time" scenarios where a leading team decelerates while the trailing team accelerates.

For continuous and mixed-domain variables, the system must deploy Gaussian and Archimedean copulas. By separating the marginal distributions (e.g., heavily skewed rushing yard totals) from the underlying dependence structure (the copula function), the engine can model intricate multivariate relationships without assuming joint normality. The integration of Copula Energy Scores (CES) will serve as the strictly proper scoring rule to evaluate these joint predictive distributions, replacing simplistic univariate energy metrics and providing the shadow-evaluation-pass.ts module with a mathematically rigorous method for A/B testing parlay pricing models.

Continuous-Time Dynamics: Survival Analysis and Hawkes Processes
The current analytical framework relies predominantly on static, snapshot-based classification and regression models, fundamentally misaligning with the continuous, sequential nature of sporting events. Treating player injuries, substitutions, and scoring drives as binary outcomes at the end of a match discards the rich temporal information embedded in the event stream. The architecture must transition to a continuous-time survival analysis paradigm, explicitly modeling these occurrences as competing risks.

The DeepHit neural network architecture provides the optimal foundation for this transition. Unlike the traditional Cox Proportional Hazards model, which assumes that covariate effects multiply the baseline hazard uniformly over time, DeepHit makes no assumptions about the underlying stochastic process. It employs a shared sub-network to learn common patient or player representations, followed by cause-specific layers that directly learn the joint distribution of survival times and event types. The loss function uniquely combines the log-likelihood of the first hitting time to manage right-censored data with a ranking loss that penalizes the incorrect temporal ordering of risks across the population. This allows the gse-ml-service to output dynamically updated hazard curves for specific events, such as the probability of a starting running back exiting the game due to injury versus tactical substitution.

To model momentum and self-excitation—phenomena often dismissed as the "hot hand" fallacy but empirically verifiable in high-tempo sequences—the engine must implement Hawkes processes. A Hawkes process is a non-Markovian point process where historical events temporarily raise the probability of future events. By utilizing a random-memory Hawkes process, where the duration of self-excitation is treated as a random variable rather than an infinite-memory exponential decay, the model can accurately capture scoring bursts and turnover cascades. This requires a multivariate excite-and-inhibit transition matrix, allowing a successful offensive play to not only increase the likelihood of subsequent offensive success but simultaneously suppress the opposing defense's hazard rate. This continuous-time intensity modeling will drastically improve the precision of the live, in-play prediction surfaces.

Decision Theory: Bipolar Kelly Ceilings and E-Processes
The application of a static, folklore-derived fractional Kelly criterion ($\kappa=0.25$) within the robust-kelly.ts module represents a fundamental misunderstanding of portfolio sizing under Knightian uncertainty. The classical Kelly criterion maximizes the asymptotic growth rate of wealth under the strict assumption that the bettor knows the true underlying probability distribution. When the model is misspecified—a certainty in complex sports markets—betting the full, or even a statically fractioned, Kelly amount guarantees severe drawdowns and eventual ruin.

To protect the sizing lockbox, the platform must implement a bipolar Kelly ceiling based on the inf-KL divergence. Under model misspecification, the optimal asymptotic growth rate is not governed by the standard Kullback-Leibler divergence between the empirical distribution and the model class. Instead, it is determined by the reverse information projection onto the bipolar enlargement of the model class, effectively testing against the set of all non-negative sequential tests (e-variables) that cannot distinguish the true distribution from the null. This inf-KL optimal rate dictates a dynamic model-class haircut. The algorithm must construct an e-process over the sequence of returns, dynamically restricting the stake size based on the observed divergence between the model's confidence set and the empirical outcomes, ensuring that the maximum drawdown remains strictly controlled.

The concept of e-processes provides the mathematical solution to a second systemic vulnerability: the invalidation of fixed-sample statistical tests through optional stopping. In an environment where founder operations and automated pipelines continuously monitor model performance, relying on standard p-values guarantees inflated false discovery rates. An e-process is a non-negative stochastic process that maintains the supermartingale property under the null hypothesis, effectively representing the wealth of a sequential bettor.

By formulating the test statistic as a sequential likelihood ratio (an e-value), the system leverages Ville's maximal inequality, which states that the probability of the e-process ever crossing a threshold $1/\alpha$ is strictly bounded by α. This property enables anytime-valid inference. The forecast-skill-eprocess.ts module can therefore continuously evaluate a new candidate model against the market consensus. If the accumulated e-process wealth crosses the threshold of 20 (for $\alpha=0.05$), the system possesses mathematically irrefutable evidence of a

predictive edge, regardless of when the evaluation is halted or how frequently the data was peeked at. This formalizes the Baxley Causal E-Process Theory (CEPT) internal hypothesis into a deployable, rigorously defended product feature.

Causal Inference and Market Microstructure

The pursuit of isolating individual player value or coaching acumen via standard regression or marginal replacement models introduces severe confounding bias, particularly due to the non-random nature of NFL scheduling and substitution patterns. To achieve unbiased causal estimands, the infrastructure must deploy Longitudinal Targeted Maximum Likelihood Estimation (LTMLE).

LTMLE is a doubly robust, semi-parametric framework designed for dynamic treatment regimes in the presence of time-varying confounders. The procedure involves estimating the conditional outcome mechanism (e.g., expected points added per play) alongside the treatment mechanism (the probability of a specific player or tactical scheme being deployed, given the historical game state). By generating Inverse Probability of Censoring Weights (IPCW) and applying a targeted fluctuation step that solves the efficient influence curve equation, LTMLE minimizes the asymptotic bias of the target parameter. If either the outcome model or the treatment model is specified correctly—facilitated by utilizing non-parametric Super Learner ensembles—the resulting causal estimate is consistent. This will transition the player props architecture from correlative heuristics to true causal effect isolation.

Simultaneously, the execution and timing of market entries require an understanding of market microstructure. The naive assumption that the posted odds represent the true efficient price ignores the mechanics of limit order books (LOB) and the shrouding of effective prices through varying vig structures. By tracking the Net Liquidity (NetLiq) velocity and the order flow imbalance on exchanges like Kalshi, the system can identify informed steam versus reactive public money.

Furthermore, the calculation of live, in-play win probabilities must incorporate the mathematics of jump-diffusion. Modeling the score differential as a biased random walk allows for the derivation of exact "safe lead" probabilities using the error function erf(L/
$4D\tau$

), where L is the lead size, D is the diffusion coefficient representing scoring variance, and τ is the remaining game time. This closed-form analytical solution provides a highly stable, computationally efficient baseline for the live predictive engine, anchoring the more volatile machine learning outputs.

Evaluation Meta-Metrics and Scoreline Frameworks

The reliance on basic accuracy and raw Brier scores obscures the underlying mechanics of model improvement or degradation. The evaluation framework must natively compute the

Murphy Decomposition of the Brier score for all probabilistic outputs, partitioning the metric into Reliability, Resolution, and Uncertainty components.

When analyzing the effects of market-anchored shrinkage (e.g., shrinking model probabilities toward the closing line by w=0.10), the Murphy Decomposition is critical. While shrinkage artificially improves Reliability by forcing the predictions to mimic the highly calibrated market, it often decimates Resolution—the model's intrinsic ability to differentiate between distinct underlying risk states. If an algorithmic update improves the aggregate Brier score solely through increased Reliability while degrading Resolution to the level of an uninformed prior, the update constitutes a loss of signal and must be rejected.

Moreover, the current architecture lacks a rigorous framework for comparing nested models. When a new feature is added to an existing model, the asymptotic distribution of standard Diebold-Mariano test statistics becomes non-standard because the forecast errors of the nested and nesting models are perfectly correlated under the null hypothesis. To prevent the acceptance of false-positive model improvements driven purely by parameter estimation noise, the system must implement the Clark-McCracken testing protocol, utilizing a bootstrap-based methodology to estimate valid finite-sample critical values for nested out-of-sample forecast comparisons.

For models predicting continuous or ordinal outcomes, such as game totals or exact scorelines, binary accuracy is a highly destructive metric. Predicting a 24-21 final score when the actual result is 24-20 contains significant predictive signal, yet binary evaluation scores it identically to a 10-3 prediction. The integration of Scoreline Partial-Credit evaluation utilizing the Continuous Ranked Probability Score (CRPS) and Energy Scores is mandatory. These proper scoring rules reward predictions that place high probability mass in the topological vicinity of the realized outcome, providing a smooth, differentiable gradient for optimizing total and margin predictors.

Data Strategy and Immutable Infrastructure
To construct these advanced architectures without violating terms of service or relying on cost-prohibitive proprietary feeds, the data ingestion pipeline must pivot toward open-source broadcast extraction. The deployment of tools akin to next-gen-scraPy enables the extraction of Multiple Object Tracking (MOT) coordinates directly from standard broadcast video using advanced computer vision and jersey-number optical character recognition. This provides high-fidelity, sub-second spatial data—essential for topological data analysis (TDA) and DeepHit survival covariates—without requiring licensing agreements with proprietary providers like Next Gen Stats (NGS).

This data will feed into the newly established shadow prediction engine documented in the repository. The shadow engine provides a persistent, stateful environment where particle filters, BAEE (Bayesian Adaptive Ensemble) weights, and Hawkes steam detectors can process real-time ingestion streams without ever publishing a live pick or contaminating the production glass ledger. To ensure the integrity of this pipeline, the infrastructure relies on scheduler-liveness.ts to diagnose silent cron failures and deploys a traffic-heartbeat.ts failsafe,

which securely hijacks organic user traffic to trigger essential ingestion runs when platform-level scheduling degrades.

Ranked Falsifiable Experiment Backlog
The following structured backlog synthesizes the theoretical proposals into a prioritized, falsifiable research program, ready for immediate execution within the Beexly/Sports repository.

ID Theoretical Hypothesis Algorithmic Method Target Data Surface Benchmark Baseline Evaluation Metric Statistical Gate Falsification Kill Line Repository Integration Path Im Rk Ef
01 Global calibration masks conditional minority collapse.
Mondrian Conformal Risk Control

Settled picks history Global Isotonic / Platt Scaling Conditional ECE & Loss Rate Kolmogorov-Smirnov Test Marginal ECE improves, but Conditional ECE within Mondrian strata degrades by >0.02. packages/prediction-engine/src/calibration-apply.ts 5 1 S
02 Independent Poisson formulations systematically misprice SGPs.
Sarmanov COM-Poisson Copula

nflverse PBP + public odds Independent Skellam / Poisson Joint Brier Score Diebold-Mariano Test Out-of-sample ΔELPD<5 or dispersion parameter v≈1.0. packages/prediction-engine/src/poisson.ts 5 3 M
03 Fixed Kelly sizing leads to overexposure under misspecification.
Bipolar inf-KL E-Process Kelly

odds_line_snapshots Fixed Fractional Kelly (κ=0.25) Return over Max Drawdown (RoMD) E-process wealth accumulation Fails to achieve ≥90% baseline growth while keeping max drawdown ≤70%. packages/prediction-engine/src/robust-kelly.ts 4 2 M
04 Nested model comparisons yield false positives due to parameter noise.
Clark-McCracken Bootstrap Test

picks-h1.json Holdout Standard Asymptotic T-test Out-of-sample MSE MSE-F Test Statistic p>0.05 using Clark-McCracken critical values on historical "wins". packages/verifier/src/ 5 1 S
05 Market shrinkage destroys model resolution for artificial reliability.
Murphy Decomposition

calibration_status Unshrunk Raw Probabilities Resolution Component Bootstrap Confidence Interval Resolution drops by >10% against an uninformed uniform prior. apps/web/lib/calibration/ 4 1 S
06 Injury and drive outcomes are continuous competing hazards.
DeepHit Survival Architecture

nflverse injuries & tracking Cox Proportional Hazards Time-dependent Concordance (C
td
) Wilcoxon Signed-Rank Test Out-of-sample C
td
<0.60 or severe CIF calibration drift. gse-ml-service/app/models/ 4 4 L
07 Scoring momentum is driven by finite-memory self-excitation.
Random-Memory Hawkes Process

PBP timestamped scoring events Homogeneous Poisson Process Sequential Log-loss
Likelihood Ratio Test ΔAIC<10 over schedule-adjusted baseline.
packages/prediction-engine/src/hawkes-steam.ts 3 3 M
08 Anytime-valid evaluation prevents p-hacking and optional stopping.
Ville's Inequality E-Processes

Live streaming prediction evaluations Fixed-N Asymptotic p-value E-value Wealth (M
t

) Crossing Time Boundary (τ) M
t

fails to permanently cross the 1/α=20 boundary.
packages/prediction-engine/src/forecast-skill-eprocess.ts 5 1 M
09 Valid player value requires adjustment for time-varying confounders.
LTMLE Causal Framework

NGS surrogate tracking / PBP Standard Expected Points Added Average Treatment
Effect (ATE) Influence Curve Variance 95% Confidence Intervals for the causal estimand
overlap 0. gse-ml-service/app/models/ 4 4 L
10 Safe lead win probabilities are analytically governed by diffusion.
Error Function (erf) Jump-Diffusion

Minute-by-minute PBP trajectories Discrete Logistic Regression In-play Log-loss Vuong
Closeness Test Sequential Log-loss reduction is <0.01 vs. empirical matching.
packages/prediction-engine/src/ 3 2 S
11 Exact score and margin models require topological evaluation.
Scoreline CRPS & Energy Scores

NFL margin and exact score history Binary Accuracy / Hit Rate Continuous Ranked
Probability Score Diebold-Mariano Test CRPS grading yields no re-ranking of models
compared to binary accuracy.packages/verifier/src/ 3 1 S
12 Broadcast video can yield license-free high-fidelity tracking.
next-gen-scraPy MOT Extraction

Public Broadcast / All-22 Feeds Proprietary NGS Data Spatial RMSE Direct Coordinate Error Coordinate translation RMSE exceeds 2.0 yards vs. truth set. packages/data-ingestion/src/ 4 4 L

(Note: Im = Impact, Rk = Risk, Ef = Effort. Scale 1-5 for Impact/Risk, S/M/L for Effort).

By executing this rigorous integration of conformal risk control, continuous-time mathematics, and e-process decision theory, Galaxy Sports Edge will not only resolve its immediate calibration inversions but permanently secure its position as the singular, mathematically irrefutable standard for predictive accountability in the industry.

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